

Exercise Sheet 7

Predicate Logic – Syntax

Note that question 4 is marked as being assessed.

- Consider the following example from lecture 11. Consider the database composed of the 3 following tables (we only show some of the rows here):

Student		Module		Enroll	
sid	name	mid	name	sid	mid
0	Alice	0	Math	0	0
1	Bob	1	OOP	1	1

These 3 tables can be seen as 3 relations:

- $Student(sid, name)$: predicate $Student$ relates student ids and names
- $Module(mid, name)$: predicate $Module$ relates module ids and names
- $Enroll(sid, mid)$: predicate $Enroll$ relates student and module ids

Write the following queries as predicate logic formulas:

- find the student names x enrolled either in the Math module or the OOP module
 - find all the pairs of student names x and y that are enrolled in the same module
- Provide an example of a domain and a signature that includes an equivalence relation, and state predicate logical formulas that capture the fact that it is an equivalence relation.
 - Define the ring laws in predicate logic. Indicate what signature you are using.
 - assessed:** Consider the following domain and signature:
 - Domain: $\mathbb{N}$
 - Functions: $0, 1, 2, \dots$ (arity 0); $+$, $\times$ (arity 2)
 - Predicates: **prime**, **even**, **odd** (arity 1); $=$, $>$, $\geq$ (arity 2)

Express the following sentences in predicate logic:

- If a number is strictly greater than another number then it must be greater than or equal to the successor of that number
 - A prime number is a number greater than 1 that cannot be expressed as the multiplication of two numbers greater than 1
- Provide a constructive Natural Deduction proof of (let's keep practicing)

$$(A \wedge \neg B) \rightarrow (A \rightarrow C) \rightarrow (D \rightarrow B) \rightarrow (\neg D \wedge C)$$